Milestone 3 Report - Geospatial Disaster Analysis

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1. Introduction

Natural disasters occur frequently and can cause significant damage to life and property. Understanding the geographical distribution of disaster declarations helps identify which regions are more vulnerable to different types of disasters. In this milestone, geospatial analysis is performed on disaster declaration data across the United States.

2. Objective

The objective of this milestone is to analyze and visualize the geographical distribution of natural disaster declarations across the United States. The analysis helps in identifying the states with the highest number of disaster declarations and understanding the most common disaster types.

3. Dataset Description

The dataset used in this project is the FEMA disaster declarations dataset. It contains information about disaster events that occurred across different states in the United States.

The dataset includes the following important attributes:

- Disaster Number
- State
- Declaration Date
- Disaster Type
- Incident Type
- Incident Begin Date
- Incident End Date

This dataset helps analyze how disasters vary across different regions.

4. Tools and Technologies Used

The following tools and technologies were used to perform the analysis:

- Python
- Jupyter Notebook
- Pandas (for data processing)
- Matplotlib (for data visualization)
- CSV dataset (cleaned FEMA disaster data)

5. Data Preparation

Before performing the analysis, the dataset was cleaned and prepared. The following preprocessing steps were performed:

- Loaded the cleaned dataset into Jupyter Notebook.
- Converted the declaration date column into datetime format.
- Checked for missing values and handled them appropriately.

- Verified important columns such as state and incident type.
- Prepared the dataset for visualization and analysis.

This preprocessing ensured that the dataset was ready for accurate analysis.

6. Data Analysis and Visualization

Two main analyses were performed in this milestone.

1. Disaster Distribution by State

The number of disaster declarations for each state was calculated using Python. The top 10 states with the highest disaster declarations were identified and visualized using a bar chart.

This visualization helps understand which states experience disasters more frequently.

2. Distribution of Disaster Types

The dataset was analyzed to determine how frequently each disaster type occurs. A bar chart was created to visualize the distribution of disaster types such as hurricanes, floods, fires, snowstorms, droughts, and earthquakes.

7. Results and Observations

The analysis revealed several important insights:

- Some states experience a significantly higher number of disaster declarations compared to others.
- Hurricanes and floods are among the most common disaster types in the dataset.
- Events such as tsunamis, volcanic eruptions, and terrorist incidents occur very rarely.

- The visualizations clearly show how disaster occurrences vary across different states.

  These insights help understand disaster patterns and regional vulnerability.

8. Conclusion

The geospatial analysis successfully identified the distribution of disaster declarations across different states in the United States. The visualizations show that certain states experience disasters more frequently and that floods and hurricanes are among the most common disaster types.